

THE ATLAS OF LOST SIGNALS

A 20-page mystery dossier for practicing PyPDFLoader

A fictional investigative report about a strange radio signal, an abandoned observatory, and a research team trying to understand a message that should not exist.

Practice goals

- Load the PDF with PyPDFLoader
- Inspect page-by-page Document objects
- Extract metadata
- Split the document into chunks
- Search for names, dates, codes, and clues
- Build a small RAG experiment

CLASSIFIED • CASE FILE 47-B

01 — The Signal

At 02:17 UTC on a cold October morning, the North Meridian Observatory detected a narrow-band radio pulse from a direction just beyond the constellation Cygnus. The signal lasted exactly 83 seconds.

The unusual part

Astronomers receive strange signals all the time. Most are interference, satellites, aircraft, atmospheric effects, or ordinary cosmic sources. This one was different because its frequency remained unnaturally stable while its intensity changed in a repeating pattern.

The waveform looked less like noise and more like someone tapping on a locked door.

The observatory automatically archived the recording as NM-47B-0217. Three independent receivers recorded the same event.

Keywords: North Meridian Observatory, 02:17 UTC, 83 seconds, NM-47B-0217

02 — The Observatory

North Meridian stands on a plateau 1,940 meters above sea level. The main dish was constructed in 1987 and upgraded twice: first in 2004 and again in 2021.

A forgotten facility

Seven kilometers east of the main station sits a smaller building known as Station Echo. It was used for experimental radio astronomy until funding disappeared in 1998.

The building has no active network connection. Its equipment is old, its generator is unreliable, and according to official records nobody has worked there for years.

Yet the morning after the signal, technician Mara Voss found a warm coffee cup inside Station Echo.

The cup was warm. The building had been locked for eleven years.

03 — The First Analysis

Dr. Elias Ren examined the raw waveform before breakfast. His first hypothesis was terrestrial interference. His second was a malfunctioning receiver.

Both were rejected after the signal appeared identically in data from three geographically separated antennas.

A repeating structure

When the waveform was converted into a spectrogram, Ren noticed seven repeating bands. The sequence was not random: 3, 1, 4, 1, 5, 9, 2.

Those digits immediately reminded the team of pi, but the connection was probably coincidence. The next sequence was 6, 5, 3, 5, 8, 9.

The team labelled the pattern **Layer A** and began searching for other layers hidden inside the recording.

04 — Mara Voss

Mara Voss had worked at North Meridian for sixteen years. She was known for solving hardware problems without documentation, often identifying a failing component simply by listening to a machine.

The notebook

Voss kept a black notebook with squared paper. On the day of the signal she wrote three sentences:

02:17. Signal repeats every 11 minutes. Echo building active.

No one told Echo to transmit.

Do not trust the clock.

The last sentence became important later. The observatory clocks were synchronized to GPS time, but Station Echo contained an old mechanical clock that ran independently.

Clue: look for references to 11 minutes, clocks, and Station Echo.

05 — The Eleven-Minute Pattern

The team replayed the recording and discovered something stranger. Every eleven minutes, a very weak echo appeared beneath the main carrier.

The echo was shifted in frequency by 1.618 kHz. Ren initially called this meaningless. Mathematician Noor Sen disagreed.

A familiar number

1.618 is close to the golden ratio, commonly written as phi. The team tested whether the delay, frequency shift, or pulse spacing contained other mathematical relationships.

They found that the strongest pulses occurred at intervals of approximately 1.618 seconds, followed by 2.618 seconds, then 4.236 seconds.

These values suggested a sequence in which each term was related to the previous two.

06 — The Hidden Text

Noor Sen transformed the signal into a binary representation by dividing each pulse into eight time slots. Long pulses were treated as 1 and short pulses as 0.

The first 64 bits produced nothing readable. Then she reversed the bit order in each byte.

The message

The decoded bytes contained fragments resembling ASCII text. Most characters were corrupted, but one sentence appeared clearly:

WHEN THE SKY IS QUIET, LISTEN BELOW.

Nobody in the room spoke for several seconds. The sentence seemed less like a scientific transmission and more like an instruction.

07 — Station Echo

At 18:40 local time, Mara Voss and Noor Sen walked to Station Echo. The door showed no signs of forced entry.

Inside, dust covered the consoles. A dead radio receiver sat beside a filing cabinet. On the wall was a faded map of the northern sky.

The second antenna

Behind the building they discovered a narrow auxiliary antenna pointed downward rather than toward the sky.

It was connected to an underground cable. The cable disappeared into a concrete conduit leading toward the plateau.

Someone had built an antenna for a signal that was not coming from above.

08 — The Underground Cable

A ground-penetrating radar survey revealed a tunnel approximately 300 meters long. It was not present on the modern site map.

The tunnel contained fiber-optic lines, copper cables, and an old emergency telephone system.

A sealed door

At the end of the tunnel was a steel door marked **MERIDIAN SUBLEVEL 3**. The observatory officially had only two underground levels.

The door required a six-digit code. A handwritten note beside the keypad contained four numbers: 11, 47, 83, and 21.

The team tried 114783. Nothing happened. They tried 478321. The lock clicked.

09 — Sublevel Three

The room beyond the door was surprisingly clean. Racks of equipment lined the walls, and a large cylindrical device occupied the center.

A monitor displayed a timer counting down from 00:11:00.

The machine

Ren recognized the device as a frequency synthesizer, but its design was decades ahead of anything listed in the observatory inventory.

A label on the machine read **PROJECT LANTERN**. Underneath it was a date: 14 June 1989.

The project had been hidden before most of the current team was born.

10 — Project Lantern

The recovered files described Project Lantern as an experiment in long-distance synchronization. Its researchers believed that certain physical systems could be used as extremely precise clocks.

The documents never mentioned aliens, extraterrestrial communication, or anything supernatural.

The dangerous idea

Instead, the project proposed transmitting timing information through a resonant underground network. The network could synchronize distant oscillators without relying on radio or satellites.

One page contained a warning: **DO NOT ACTIVATE DURING LOW-SIGNAL INTERVALS.**

The team did not know what a low-signal interval meant.

11 — The Missing Researcher

Project Lantern had three principal researchers: Helena Ward, Arun Patel, and Viktor Hale.

Ward retired in 1997. Patel moved to another country. Hale disappeared from institutional records in 1993.

A photograph

In an archived personnel photograph, Hale stood beside the same cylindrical machine now sitting in Sublevel Three.

On the back of the photograph was a handwritten line:

If Lantern wakes again, do not answer it.

No one knew who had written the warning.

12 — The Clock

The mechanical clock inside Station Echo was found running exactly eleven minutes behind GPS time.

This explained Mara Voss's note: **Do not trust the clock.** But it raised a new question. Why eleven minutes?

A timing experiment

When the team synchronized the mechanical clock with the signal, the weak echo disappeared. When they moved the clock eleven minutes backward, the echo returned.

The clock was not measuring time incorrectly. It was apparently acting as part of the experiment.

The clock was a key, not a clock.

13 — The Second Message

The team waited for the next major pulse. At 02:17 the following morning, the signal returned.

This time the decoded text was longer.

WHEN THE SKY IS QUIET, LISTEN BELOW. ELEVEN MINUTES. THIRD LEVEL. DO NOT SEND A REPLY.

The final instruction was unmistakable. Whatever produced the message expected someone to respond.

Ren argued that silence was the safest option. Sen argued that the system might already be responding automatically.

Mara Voss simply unplugged the transmitter.

14 — The Unplugged Transmitter

For six minutes nothing happened. Then the sublevel machine powered itself on.

Its internal battery was supposed to have been removed in 1998.

The impossible reading

The display showed an incoming frequency, but no antenna was connected. The signal was apparently entering through the building's electrical grounding system.

The machine began generating the same seven-band pattern observed in the original recording.

The transmitter was not receiving power. It was receiving a signal.

15 — The Map

Noor returned to the sky map in Station Echo. Seven stars had been marked with red pencil.

The stars formed a rough geometric pattern. When connected, they pointed toward a location on Earth: an isolated valley approximately 410 kilometers north-east of the observatory.

Coordinates

A note beneath the map gave coordinates in an outdated format: 45° 17' 09" N, 91° 42' 31" W.

The location was not a city, military base, or research station. Satellite imagery showed only forest and a narrow river.

16 — The Valley

A field expedition reached the valley three days later. Beneath the trees they found a concrete foundation partially covered by moss.

The structure contained no obvious machinery. However, a circular depression in the ground matched the dimensions of the Project Lantern cylinder.

The final clue

Carved into the concrete was a sequence of symbols. The first seven matched the seven frequency bands from the signal.

The eighth symbol was different. It resembled an open eye.

The team had been looking for the source of the signal. The valley suggested the signal was looking for them.

17 — The Choice

Back at North Meridian, the team faced a decision. Project Lantern could be permanently dismantled, or they could investigate its purpose.

Ren wanted destruction. Sen wanted controlled observation. Voss proposed a third option: build a passive receiver and never transmit.

The experiment

They built the receiver in an isolated room and connected it to no network, no transmitter, and no external power source beyond a shielded battery.

At 02:17, the receiver produced a single sentence.

WE KNOW YOU ARE LISTENING.

18 — The Archive

The team preserved every recording, photograph, notebook, and instrument log. They created a digital archive called CASE 47-B.

What remains unexplained

No known satellite crossed the relevant part of the sky during the original transmission. The underground system was omitted from official construction plans. Project Lantern's power source remains unknown.

Most importantly, the signal contained information that appeared to depend on decisions made by the research team after the signal had already been detected.

The strongest evidence was not what the signal said. It was what it seemed to know.

19 — Research Notes

The case generated several competing explanations.

Hypothesis A — Hidden terrestrial transmitter

A secret transmitter could explain the signal, the underground cables, and Project Lantern. It does not easily explain the apparent prediction of later events.

Hypothesis B — Experimental timing phenomenon

Lantern may have created a previously unknown synchronization effect. The messages could be artifacts of a system designed to encode timing information.

Hypothesis C — Unknown origin

The least testable explanation is that the signal originated outside the known system. The evidence is intriguing but insufficient to prove this.

Practice tip: this page is useful for semantic search because it contains repeated keywords and contrasting hypotheses.

20 — Case Closed?

The North Meridian Observatory officially classified the incident as an unresolved equipment anomaly.

The public report mentions no underground level, no Project Lantern, and no missing researcher.

The final entry

Mara Voss's notebook ends with a single line dated eleven days after the first signal:

02:17 again. This time the message was only three words.

WE ARE EARLY.

The notebook was closed. The case was archived. The receiver was placed in a locked cabinet.

Whether the Atlas of Lost Signals describes an experiment, an elaborate deception, or something genuinely unknown is left to the reader.

End of Case File 47-B

For PyPDFLoader practice, try extracting every page and searching for: **Lantern, Voss, 02:17, eleven, 47-B, third level, WE KNOW YOU ARE LISTENING.**